Forging the Future: Market Intelligence and Growth Strategy for High-Performance Metals Through 2035

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Table of Contents

1. Introduction 3
2. Report Objective and Scope 3
3. Introduction to the Analytical Framework and Methodology 4
4. Metal Markets: Production & Form 5
4.1 Nickel 6
4.2 Cobalt 6
4.3 Chromium 7
4.4 Molybdenum 7
4.5 Titanium 7
4.6 Tungsten 8
4.7 Other Selected Metals 8
5. Industry Needs 9
5.1 Automotive 9
5.2 Oil and Gas 11
5.3 Energy and Environmental Technology 12
5.4 Aerospace 13
5.5 Chemical Processing 15
5.6 Electronics and Electrical Engineering 16
5.7 Medical Technology 17
6. Global Market Dynamics and Trade 19
7. Market Scenarios to 2035 20
7.1 Market Model and Scenario Assumptions 20
7.2 Results by Industry, Product, and Region 21
8. Strategic Recommendations 22
8.1 Priority Products, Industries, Regions, and Customers 22
8.2 Recommended Actions and Implementation Roadmap 24
9. Conclusion 25

References 26
Appendices: Data, Sources, Calculations, and Assumptions 40
A1 Definitions and the stage ladder 40
A2 Reproducible scenario calculations 40
A3 Data sources, product codes and reproducibility 40
A4 Information needed before investment 40

1. Introduction

For centuries, metals have been among the essential materials on which human societies are built, enabling the construction of cities, transport systems, power networks, machinery and technologies that support modern economies. Their uses have expanded with each wave of industrial development, and today they remain central to established industries as well as emerging technologies. This enduring role is reflected in the sector's economic scale: according to McKinsey & Company's Global Materials Perspective, the global metals and mining industry generated approximately US$3.0 trillion in revenue in 2024, after a decline of 6 percent, and grew by only about 1 percent in 2025 (McKinsey & Company, 2025, 2026).

Within this broad market, high-performance alloys serve applications where ordinary materials may not meet the required operating conditions. These engineered materials combine metals such as nickel, cobalt, titanium, iron and chromium to provide properties such as strength at high temperatures, corrosion resistance, oxidation resistance or reliable performance under pressure. Nickel- and cobalt-based superalloys, titanium alloys and corrosion-resistant specialty steels are examples. Demand comes from aerospace and defence, power generation, chemical processing, oil and gas, medical technology and other industries with demanding technical requirements. The production and availability of metals including nickel, cobalt, chromium, molybdenum, titanium, tungsten, zirconium, copper, aluminium and manganese therefore provide important context for understanding this specialist market.

The outlook for high-performance alloys to 2035 is positive, although growth is likely to differ across alloy types and applications. One market estimate places the segment at US$13.6 billion in 2025 and projects it to reach US$19.0 billion by 2034 (The Insight Partners, 2025), although other providers start as low as US$10.2 billion because they draw the boundary of the segment differently (IMARC Group, 2025), while broader industry analysis points to continued demand from aerospace, defence, energy and other industrial uses. This growth will depend on the pace of investment and technology adoption, as well as the cost and availability of alloying metals. The strongest prospects are likely to be in applications where a particular combination of material properties, reliability and qualification requirements creates lasting value for customers.

2. Report Objective and Scope

The objective is to identify where changes in metal supply, trade and industrial demand create commercially relevant opportunities for Münvest through 2035. The central research question is: How do the production, availability, trade and industrial uses of selected metals shape opportunities in high-performance materials?

The report covers ten named metals and a residual category for other materials. Its industrial scope comprises automotive, oil and gas, energy and environmental technology, aerospace, chemical processing, electronics and electrical engineering, and medical technology. The metal chapters provide market context, and the industry chapters translate that context into applications, product forms and buyers. VDM Metals' published portfolio serves as an application benchmark, not as proof of Münvest's own production capability, customer relationships or market share.

The analysis combines structured statistics from USGS, UN Comtrade, Eurostat PRODCOM, the World Bank, OECD and UNIDO with industry outlooks and the filings of 61 producers, suppliers and customers. Every figure keeps its observation year, unit and production stage, so that like is only compared with like. Global metal production, alloy melting output, finished-product shipments and industry revenue have different boundaries and are not interchangeable measures of market size. No aggregate addressable-market value is asserted where comparable data are unavailable.

The scenarios are sensitivities for relevant application demand, not forecasts of metal prices, company revenue or investment returns. Internal sales, margins, approvals, capacity and customer information are required before converting market findings into a company business plan.

3. Introduction to the Analytical Framework and Methodology

Because the scope covers ten metals and seven industries, the report needs one lens that makes markets with different units, stages and buyers comparable, and the team therefore built its own metal centred value chain framework. It follows each metal along four dimensions, Supply, Trade, Demand and Outlook, and only then asks what the combined picture means for the company. The sequence is deliberately metal first and industry second, because an alloy producer earns its margin by converting scarce, often concentrated raw materials into qualified products, and an opportunity is only real when the material can be sourced, moved, sold and defended over a ten year horizon.

The four dimensions answer four different management questions, and Table 1 shows how each one is measured and where it appears in the report, so that a reader can trace every conclusion back to its evidence.

Table 1: The metal centred value chain framework

Dimension	Management question	What is measured	Main evidence	Chapter
Supply	Can the metal be sourced, in which form and from whom?	Production by stage (ore, refined or intermediate, alloy melt, mill product), producer and country concentration, prices and policy constraints	USGS, study groups (INSG, ICSG, Cobalt Institute, IMOA, ICDA), World Bank Pink Sheet, producer reports	4
Trade	Where does the material move, and who depends on whom?	Exports and imports by product code, partner and year, unit values, sourcing dependence	UN Comtrade, Eurostat PRODCOM and Comext, OECD ICIO	6
Demand	Which industries need which alloy, in which product form, and who buys it?	Industry activity, components, materials, delivery forms, named buyers and competitors	OICA, ACEA, Airbus, Boeing, IEA, Cefic, SEMI, WSTS, OECD Health, company filings	5
Outlook	How large could relevant demand be in 2035, and under which assumptions?	Indexed demand scenarios by industry, with explicit growth assumptions	Sector evidence translated into transparent rates	7

Source: team framework, evidence base as listed in the References.

The discipline that holds the framework together is the stage ladder, which is the rule that a figure is only compared with a figure from the same stage of the value chain. Each metal is followed from mine output to refined metal or intermediate, then to alloy melt, mill product and finally component, and the report never adds tonnes from different rungs. Contained metal is kept apart from gross weight, mine output from refined output, actual output from capacity, and a commodity benchmark from the selling price of a qualified alloy product. This rule explains why chapter 4 reports nickel mine output, primary nickel and stainless melt on separate lines, and why the report does not claim one aggregate addressable market value where the underlying series are not comparable.

Within that discipline every metal profile in chapter 4 follows the same five questions, so that a manager can read the ten metals side by side. The profile states the commercial forms in which the metal is traded, the production stage and its geographic concentration, the price benchmark and what it does and does not measure, the availability risk created by quotas, export controls or byproduct dependence, and finally the relevance for the company's own product forms of strip, sheet, plate, bar and wire. The same structure is summarised in the scorecard that opens chapter 4.

The framework is applied in a fixed order, which is what turns analysis into a recommendation. Supply and Trade establish whether the company can secure and move the material, Demand establishes which industries pay for its conversion, and Outlook tests how resilient that demand is under slower or faster conditions. Chapter 8 then integrates the four dimensions and assesses each opportunity on eight criteria, namely market attractiveness, growth to 2035, strategic fit with the portfolio, competitive intensity, technology attractiveness, customer attractiveness, entry barriers and the potential for differentiation. In data terms the dimensions cover one chain, because Supply describes the

material and its product form, Demand the application and industry, Trade the geography, and Outlook the market size and growth.

The evidence base follows a fixed hierarchy, and that hierarchy determines how much weight a number carries. Structured statistics come first and were pulled directly, with every query logged, from UN Comtrade, Eurostat PRODCOM, the World Bank Pink Sheet, the USGS Mineral Commodity Summaries 2026, OECD STAN and ICIO, and UNIDO INDSTAT. Industry associations and official outlooks such as the IEA, Airbus, Boeing, OICA, Cefic and SEMI come second, and the annual reports and filings of 61 producers, customers and suppliers come third. Commercial market reports are used only to triangulate, every key figure is checked against a second independent source, and a missing observation stays missing rather than being estimated, which is the standard the following chapters apply to each metal in turn.

4. Metal Markets: Production & Form

The Supply dimension starts with the metals themselves, because every alloy the company sells is a conversion of raw materials whose availability, form and price it does not control. Each profile below separates the physical metal market from the qualified products used downstream and states which rung of the stage ladder a figure describes. Table 2 condenses the ten profiles into one scorecard, so that the reader sees at a glance where supply is concentrated, where prices move and where the company's own product forms are directly exposed.

Table 2: Metal market scorecard, 2025

Metal	World output 2025 (stage, unit)	Largest producer	2025 price signal	Availability risk to 2035	Relevance for the company
Nickel	3.90 Mt contained, mine	Indonesia, about 67%	USD 15,162/t, down 41% from 2022	Medium: Indonesian ore quotas	Core: base of all nickel alloys and FeNi strip
Cobalt	310 kt contained, mine	DR Congo, about 73%	Metal up 130% in 2025	High: Congolese export quota	High: superalloys, CoCr medical bar and powder
Chromium	51 Mt gross, chromite ore	South Africa, 45%	Ferrochrome led by China	Medium: smelting concentrated in China	High: alloying element in every corrosion alloy
Molybdenum	260 kt contained, mine	China	USD 51/kg oxide, up 7%	Low to medium: copper byproduct	High: C-276, 59 and 625 grades
Titanium	370 kt sponge	China, 70%	USD 12/kg U.S. import value	High for aerospace grade sponge	Indirect: substitute and complement
Tungsten	85 kt contained, mine	China, 79%	Rising, export licences	High: Chinese quotas and licences	Low: minor alloying element
Zirconium	1.2 Mt gross, concentrates	Australia	USD 1,800/t premium zircon	Medium: nuclear grade sponge is narrow	Selective: chemical and nuclear Zr products
Copper	23.2 Mt mine, 28.6 Mt refined	Chile (mine), China (refined)	USD 9,947/t, record in 2026	Medium: 25% primary gap by 2035	Moderate: CuNi and copper strip
Aluminium	74 Mt primary	China, 61%	USD 2,632/t	Low for the company	Low: competing material
Manganese	20 Mt contained, mine	South Africa	USD 4.5 per mtu ore	Low	Low: alloying addition

Sources: USGS Mineral Commodity Summaries 2026 commodity chapters, World Bank Pink Sheet, Cobalt Institute, ICSG, IAI and IMOA as cited in sections 4.1 to 4.7. Shares are team calculations on rounded USGS totals.

The scorecard already points to the central finding of this chapter, which is that the company's raw material risk sits in cobalt, in Indonesian nickel ore policy and in aerospace grade titanium sponge, while copper and aluminium matter mainly as competing or complementary materials. Nickel, as the base of the whole portfolio, is therefore the metal that is analysed first.

4.1 Nickel

Nickel is the base of the company's portfolio, and its supply has moved decisively to one country within a decade. World mine production of contained nickel reached an estimated 3.90 Mt in 2025, of which Indonesia supplied about 2.60 Mt, or roughly two thirds (U.S. Geological Survey, 2026o, 2026p). Mine output is not the metal a mill buys, however, because primary nickel is a later stage, and the International Nickel Study Group puts world primary production at 3.875 Mt in 2025 against usage of 3.577 Mt (International Nickel Study Group, 2026a, 2026b). Just over half of primary supply in China and Indonesia is nickel pig iron, which feeds stainless steel and is not the class 1 metal that vacuum melted alloys require (International Nickel Study Group, 2025).

That oversupply explains the price, and the price explains why raw material cost is not the company's main constraint. The World Bank nickel price averaged USD 25,834 per tonne in 2022 and USD 15,162 in 2025, a fall of about 41 percent (World Bank, 2026a, 2026b). The study group reported a surplus of 299 kt in 2025 but cut its 2026 surplus forecast to only 19 kt, because Indonesia reduced its ore mining quota (International Nickel Study Group, 2026c, 2026d). The initial 2026 quota of 260 to 270 Mt of ore was later revised to 290 to 300 Mt, which shows that Jakarta now manages the price through volume (Argus Media, 2026; Indonesia Business Post, 2026).

For the company the relevant market starts after the commodity, where conversion, qualification and product form create value. Stainless steel remains the first use of nickel, with world melt shop output of 64.2 Mt in 2025, of which the European Union produced 5.7 Mt (worldstainless, 2026a). German exports of nickel plate, sheet and strip (HS 7506) reached 17,775 t worth USD 578 million in 2024, an average unit value near USD 32,500 per tonne, about twice the exchange price, and Eurostat records the same product family within 1.8 percent (Eurostat, 2026a; United Nations, 2026). Competitors confirm the scale of the qualified segment, as Acerinox's High Performance Alloys division, which combines VDM Metals and Haynes, melted 83 kt in 2025 and sold EUR 1,683 million (Acerinox, 2026c), while ATI consumed about 70 million pounds of nickel and passes most price moves through surcharges (ATI Inc., 2026a).

The implication is that nickel availability is secure in volume but not in grade, so the company should secure class 1 metal and qualified revert under surcharge contracts rather than take price positions. Cobalt, the second superalloy element, shows the opposite pattern of scarcity created by policy.

4.2 Cobalt

Cobalt matters to the company through superalloys and cobalt chromium medical alloys, and its supply is the most concentrated of all ten metals. World mine production was an estimated 310 kt of contained cobalt in 2025, of which the Democratic Republic of the Congo produced about 230 kt, or 73 percent (U.S. Geological Survey, 2026h, 2026i). Refining is even more concentrated, because China produced 78.6 percent of the 240 kt of refined cobalt in 2025 (Cobalt Institute, 2026a, 2026b).

Demand is dominated by batteries, which makes alloy buyers a small customer in a large chemical market. Electric vehicle batteries took 113 kt and portable electronics 93 kt in 2025, while superalloys took only 20 kt (Cobalt Institute, 2026c, 2026d, 2026e). This small share is nevertheless the most demanding one, since aerospace requires alloy grade metal from a handful of qualified refineries.

Availability of cobalt is now set by government policy in Kinshasa rather than by geology. Congo banned exports in February 2025 and replaced the ban with a quota of up to 96,600 t a year for 2026 and 2027 (U.S. Geological Survey, 2026j). CMOC produced 117,549 t of cobalt in 2025 but holds an export entitlement of only 31,200 t for 2026 (CMOC Group, 2026a, 2026b). Prices reacted sharply, as cobalt metal in Europe rose from USD 10.75 to 24.75 per pound during 2025, an increase of 130 percent (Cobalt Institute, 2026f).

For the company this means that cobalt is a cost and continuity risk rather than a growth market, which calls for multi source contracts with Western refiners and revert recovery from aerospace customers. Chromium, the next element, is less scarce but equally dependent on one smelting country.

4.3 Chromium

Chromium is present in every corrosion and heat resistant alloy the company sells, and its value chain separates mining from smelting. World chromite ore output was an estimated 51 Mt gross weight in 2025, of which South Africa mined about 23 Mt (U.S. Geological Survey, 2026e, 2026f). These are ore tonnes, not contained chromium, and the alloy industry buys ferrochrome or chromium metal, which are later stages.

Smelting has moved to China although China holds almost no ore, which creates a dependence one step downstream of the mine. China produced 9.9 Mt of ferrochrome in 2025, while South Africa and Zimbabwe together fell 48 percent to 2.1 Mt (International Chromium Development Association, n.d.). The demand pool is stainless steel, with 64.2 Mt of melt shop output in 2025, and the United States imports 79 percent of its chromium needs (U.S. Geological Survey, 2026g; worldstainless, 2026b).

For a European producer the only integrated alternative is Outokumpu, which runs the Kemi mine and the Tornio ferrochrome plant in Finland and earned EUR 138 million of adjusted EBITDA in ferrochrome in 2025 (Outokumpu, 2026). The company should therefore treat chromium metal and low carbon ferrochrome as qualified inputs that need a second European or Japanese source. Molybdenum, the next element, is the key addition for the most corrosion resistant grades.

4.4 Molybdenum

Molybdenum separates ordinary stainless steel from the corrosion alloys that earn the highest premiums, and its intensity differs by two orders of magnitude. One tonne of 316L contains about 21 kg of molybdenum, whereas Alloy C-276 contains about 16 percent and Alloy 625 about 9 percent (ATI Inc., n.d.; Haynes International, 2023b). A small volume of nickel alloy therefore carries a large molybdenum bill.

Molybdenum supply is moderately concentrated, and a large part of it is a byproduct of copper mining. USGS estimates world mine output at 260 kt of contained molybdenum in 2025, while the International Molybdenum Association reports about 305 kt, a gap that sits almost entirely in China (International Molybdenum Association, 2026a; U.S. Geological Survey, 2026l, 2026m). Only China and the United States mine molybdenum both as a primary product and as a byproduct, so most supply follows the copper mine plan rather than the molybdenum price. The U.S. oxide price rose 7 percent to an estimated USD 51 per kg in 2025 (U.S. Geological Survey, 2026n).

Europe remains the second largest user, although its use fell 7 percent in 2025, which reflects the weak chemical cycle described in chapter 5 (International Molybdenum Association, 2026b). For the company molybdenum is a pass through cost under surcharges, and its price links the profitability of the C-276 and Alloy 59 product lines to copper mining. Titanium, the next metal, is less an input than a competitor and partner material.

4.5 Titanium

Titanium matters to the company as a substitute for nickel alloys in lighter parts and as a complementary material that the same aerospace buyers qualify. Mineral concentrates and titanium metal are different markets, since world output of ilmenite and rutile was 9.8 Mt of TiO2 in 2025, mostly for pigment, while sponge metal reached only 370 kt (U.S. Geological Survey, 2026q, 2026u).

Sponge supply is concentrated in China, which produced about 260 kt and holds 320 kt of the 470 kt of world capacity (U.S. Geological Survey, 2026r, 2026s). Chinese sponge is not the qualified aerospace source for Western engines, however, and the United States has no operating sponge capacity at all (U.S. Geological Survey, 2026t). Osaka Titanium is raising aircraft grade capacity from 40 kt to 50 kt by fiscal 2027, and ATI quotes lead times of about 20 months for premium titanium (OSAKA Titanium technologies, 2025).

Europe melts and fabricates titanium without producing sponge, with EU sold production of wrought titanium of 22.6 kt in 2024 against imports of 31.7 kt (Eurostat, 2024). The lesson for the company is that aerospace buyers face the same qualified capacity shortage in titanium as in nickel superalloys, which strengthens the bargaining position of every qualified melter. Tungsten, the next metal, shows how export controls can reshape a market within one year.

4.6 Tungsten

Tungsten is a minor alloying element for the company, but it is the clearest case of a metal whose supply is controlled by one state. World mine output was an estimated 85 kt of contained tungsten in 2025, of which China produced about 67 kt, or 79 percent (U.S. Geological Survey, 2026v). China sets national mining quotas, with a first batch of 58,000 t of concentrate for 2025, and since February 2025 requires export licences for tungsten intermediates, carbide and certain tungsten nickel alloys (Ministry of Commerce, 2025; Ministry of Natural Resources, 2025).

Most tungsten is used as carbide, which takes 65 percent of first use, while steels and superalloys take 14 percent (International Tungsten Industry Association, n.d.). European producers respond by recycling, as Sandvik reported that its powder business more than doubled orders and revenues on higher tungsten prices in the third quarter of 2026 (Sandvik, 2026). For the company tungsten is a compliance and sourcing topic for a few superalloy grades rather than a market, and the same is true for the other metals in section 4.7, which are analysed with a lighter profile.

4.7 Other Selected Metals

The remaining metals are relevant either as niche product lines or as competing materials, so their profiles concentrate on what changes the company's position.

4.7.1 Zirconium

Zirconium illustrates why mineral output and specialty metal must be kept apart. World production of zirconium mineral concentrates was about 1.2 Mt gross weight in 2025, led by Australia with 400 kt, and its main uses are ceramics, foundry sand and refractories (U.S. Geological Survey, 2026b, 2026c). Premium zircon sold at about USD 1,800 per tonne in 2025, and Iluka received USD 1,514 per tonne in the first half of 2026 (Iluka Resources Limited, 2026; U.S. Geological Survey, 2026w). The metal market that matters to the company is much smaller and is driven by nuclear fuel cladding and chemical equipment. There are 443 operable reactors and 81 under construction worldwide, and installed nuclear capacity rises from 420 GW in 2024 to 710 GW in 2035 in the IEA Net Zero Scenario (International Energy Agency, 2025h; World Nuclear Association, 2026a). Reactor grade zirconium is a narrow, qualified business held by a few producers such as ATI, which makes it attractive only where the company already has approvals (ATI Inc., 2026b).

4.7.2 Copper

Copper connects the company to electrification, but the relevant product is a small, specialised part of a large metal market. World mine production was about 23.2 Mt in 2025 and refined production 28.6 Mt (International Copper Study Group, 2026a, 2026b). Flat rolled plate, sheet and strip account for only 11 percent of first use of copper semis, against 63 percent for wire (International Copper Study Group, 2025). The copper price averaged USD 9,947 per tonne in 2025 and set records in 2026, while the IEA expects a primary supply deficit of about 25 percent by 2035 (International Energy Agency, 2026e; World Bank, 2026c). Data centres illustrate the new demand, since NVIDIA states that a 1 MW rack at 54 VDC needs up to 200 kg of copper busbar (NVIDIA, 2025). The company's opportunity

lies in copper nickel and specialty strip, where Aurubis, Wieland and Mitsubishi Materials are established, rather than in tonnage copper (Aurubis, 2025).

4.7.3 Aluminium

Aluminium is mainly a competing material for the company, because it replaces heavier alloys wherever temperature and corrosion allow it. World primary production reached about 74 Mt in 2025, of which China produced 61 percent, and the price averaged USD 2,632 per tonne (International Aluminium Institute, 2026; U.S. Geological Survey, 2026d; World Bank, 2026d). European primary production has fallen to about 955 kt, against demand of about 9 Mt a year (European Aluminium, n.d.; Federation of Aluminium Consumers in Europe, 2024). The strategic question for the company is therefore substitution in heat exchangers and busbars, not entry into aluminium production.

4.7.4 Manganese

Manganese is an alloying addition whose tonnage is driven by carbon steel. World mine output was about 20 Mt of contained manganese in 2025, led by South Africa, Gabon, Ghana and Australia, which together supplied about 81 percent (International Manganese Institute, 2026; U.S. Geological Survey, 2026k). The ore price fell to about USD 4.5 per metric ton unit, so manganese is neither a supply risk nor a growth market for the company.

4.7.5 Additional Metals

Other metals enter the portfolio only through specific applications, so this category stays a scoped extension rather than a residual total. Iron is the most important, because FeNi alloys such as FeNi36 for display masks and FeNi42 for lead frames are among the company's precision products, and their value is set by rolling capability rather than by iron supply. Together with the metal profiles above, these supply facts frame the industry analysis that follows, which asks where each metal turns into a paying application.

5. Industry Needs

The Demand dimension asks where the metals of chapter 4 turn into paying applications, and it answers that question industry by industry along the chain from demand driver to component, material, product form and buyer. Industry revenue indicates economic activity, not addressable alloy demand, so each section uses the most direct measurable indicator available, such as engine deliveries, gas turbine capacity, implant procedures or fab equipment spending. Named companies are prospects unless a relationship is documented, and VDM Metals' published applications serve as a benchmark of what a German producer can sell, not as evidence of the company's own approvals. To avoid double counting, electrolysers and fuel cells are assigned to energy, process vessels to chemicals, and sensors and magnetic components to electronics, and the seven sections then feed the scenarios in chapter 7.

5.1 Automotive

Automotive is a large but shrinking market for the company, because most of its car alloys go into engine and exhaust parts that electrification removes. The remaining value sits in hybrids and North American gasoline engines in the medium term and in soft magnetic alloys for electric drives in the long term.

5.1.1 Market Size and Electrification to 2035

The world built 96.4 million motor vehicles in 2025, up from 92.7 million in 2024, and China alone built 34.5 million, or about 36 percent (International Organization of Motor Vehicle Manufacturers, 2026a, 2026b). Europe is not growing in volume, since EU car registrations of 10.8 million in 2025 were still 17 percent below 2019 (European Automobile Manufacturers' Association, 2026). Electric cars, meaning battery electric and plug in hybrid models, exceeded 20 million sales in 2025 and reached 25 percent of new car sales (International Energy Agency, 2026f, 2026g).

Table 3: Automotive market indicators, 2025 and 2035

Indicator	2025	2035	Source basis
World motor vehicle production	96.4 million	No official forecast	OICA
Electric share of new car sales, world	25%	About 50%	IEA, both exploratory scenarios
Electric share of car sales, China	Close to 55%	Above 90%	IEA
Electric share of light duty sales, EU	26.8% of registrations (BEV plus PHEV)	Above 90%	ACEA, IEA Stated Policies
Turbocharger production	Close to 50 million units	Back toward 2022 level by 2030	Garrett citing S&P Global Mobility

Sources: OICA (2026), ACEA (2026), IEA Global EV Outlook 2026, Garrett Motion 10-K 2025 (Garrett Motion Inc., 2026a; International Energy Agency, 2026h, 2026i, 2026j; International Organization of Motor Vehicle Manufacturers, 2026a).

The regional split matters more than the world average, because electric cars exceed 90 percent of sales in China and the EU by 2035 while internal combustion sales never return to their 2017 peak in any IEA scenario (International Energy Agency, 2026h). Lithium iron phosphate, which contains no nickel, already took over 55 percent of electric vehicle batteries in 2025, so even the battery does not compensate for lost engine alloy (International Energy Agency, 2026k).

5.1.2 Key Applications and Opportunities for the Company

The company's car alloys serve turbochargers, exhaust sensors, spark and glow plugs, gaskets and catalyst foil, and eight of its eleven benchmark applications depend on the engine. Garrett names a nickel, iron and chromium alloy among its main raw materials and expects gasoline turbo penetration to peak at 54 percent in 2032, while BorgWarner won three gasoline turbocharger programmes in October 2026 with production from 2027 (BorgWarner Inc., 2026; Garrett Motion Inc., 2026b).

Table 4: The company's automotive applications, outlook to 2035

Outlook	Applications	Materials and product forms
Grows	Resolvers, current sensors and magnetic shielding in electric drives	Soft magnetic FeNi strip
Stable to 2030	Turbocharger parts for hybrids and gasoline engines, airbag parts	Heat resistant nickel alloy bar and strip
Falls	Glow plugs, diesel exhaust parts, valves and fuel injectors	Nickel alloys, heat resistant stainless strip

Source: VDM Metals application benchmark, outlook is the team's assessment.

The buyers with filed evidence are Garrett and BorgWarner, and the competing mills are Aperam Alloys Imphy in FeNi strip and Special Metals and Carpenter in engine alloys. The profit pool is shrinking rather than moving, which is why automotive should finance the company's growth areas rather than receive new capacity.

What the company should do:

- Keep supplying turbocharger, exhaust and sensor makers through 2030, with priority for North American gasoline and hybrid platforms.
- Grow soft magnetic FeNi strip for electric drive resolvers and current sensors with European Tier 1 suppliers.

- Exit diesel only parts such as glow plugs first, because their volume falls fastest after 2027.

5.2 Oil and Gas

Oil and gas is a stable market for the company, in which spending holds its level to 2035 while its mix moves from oil to gas and from the Middle East toward the Americas and Africa. The company's advantage is that many of its alloys are approved for sour gas service, which keeps most competitors out.

5.2.1 Investment, Gas and LNG to 2035

Upstream oil and gas investment was just under USD 570 billion in 2025 and about USD 546 billion in 2026, with gas up 8 percent and oil slightly down (International Energy Agency, 2025i, 2026r). About 40 percent of upstream spending only slows the decline of existing fields, which supports replacement and workover demand for corrosion resistant parts (International Energy Agency, 2025g).

Table 5: Oil and gas indicators, today and 2035

Indicator	Today	2035	Source basis
Upstream oil and gas investment	USD 570 bn (2025), USD 546 bn (2026)	Near today's level (Stated Policies), about 10% higher (Current Policies)	IEA
Oil demand	100 mb/d (2024)	Peak near 102 mb/d around 2030, back near 100 mb/d	IEA Stated Policies
LNG trade	560 bcm (2024)	880 bcm (Current Policies)	IEA
Upstream growth regions 2026	Africa +12%, Central and South America +10%	Continued shift	IEA

Sources: IEA World Energy Outlook 2025 and World Energy Investment 2026 (International Energy Agency, 2025i, 2025j, 2025k, 2025l, 2026s).

Geography is shifting, because upstream investment rises by over 10 percent in Central and South America and in Africa in 2026, while the Middle East falls slightly and Qatar's North Field East start up slips to at least the first half of 2027 (International Energy Agency, 2026q, 2026s).

5.2.2 Key Applications and Opportunities for the Company

The company sells nickel alloy bar, plate, strip and welding wire for wells, pipes and LNG equipment, and the sour service standard ISO 15156 is its main barrier to entry. Clad pipe is the growing form, because it uses a thin nickel alloy liner on carbon steel, and BUTTING, the leading clad pipe maker, reports turnover of EUR 880 million (BUTTING Group, 2026).

Table 6: The company's oil and gas applications, outlook to 2035

Outlook	Applications	Product forms
Grows	Clad and lined pipe, subsea tubes and umbilicals, LNG cryogenic parts	Strip, plate, welding wire, FeNi36
Stable	Well tools, OCTG accessories, flanges and fittings, sour gas processing	Bar, billet, plate
Falls	Oil only refinery revamps in Europe	Plate

Source: VDM Metals application benchmark, outlook is the team's assessment.

Demand is visible in the order books of the equipment makers, as TechnipFMC reported subsea orders of USD 11.2 billion and a backlog of USD 16.6 billion for 2025 (TechnipFMC plc, 2026). Competitors are investing at the same

time, since Alleima started a fourth VAR furnace at Sandviken in 2025 and Outokumpu is adding about 13,000 t of high nickel flat capacity at Avesta from 2028 (Alleima AB, 2026b; Outokumpu Oyj, 2026). Acerinox's alloy division, which competes most directly with the company, called oil and gas sluggish in 2025, which shows that the segment rewards approvals rather than volume (Acerinox, 2026d).

What the company should do:

- Focus sales on gas and offshore projects in Central and South America and in Sub Saharan Africa, where upstream spending grows fastest.
- Sell more strip and welding wire for clad pipe through BUTTING and similar pipe makers, protected by sour service approvals.
- Defend bar for well tools with SLB and TechnipFMC through framework agreements rather than single project bids.

5.3 Energy and Environmental Technology

Energy and environmental technology contains the company's fastest growing near term market, gas turbines, next to a hydrogen market whose orders have collapsed. The opportunity is therefore component specific, and growth in solar, wind or electricity consumption cannot be translated directly into nickel alloy demand.

5.3.1 Gas Turbines, Nuclear and Hydrogen in Numbers

Gas turbines are the tightest pocket, because data centres and grid stability have created more orders than the industry can build. Siemens Energy told investors in November 2025 that its gas turbine capacity is sold out until fiscal 2028 and will grow from about 17 GW in fiscal 2024 to above 30 GW a year from fiscal 2028 (Siemens Energy, 2025). The binding constraint is hot section castings and forgings made from nickel base superalloys, not nickel ore.

Table 7: Energy technology indicators relevant to high performance metals

Indicator	Latest value	Direction to 2035	Metal and product form
Siemens Energy gas turbine capacity	About 17 GW (FY2024), sold out to FY2028	Above 30 GW a year from FY2028	Superalloy bar, billet and forging stock
Reactors under construction	81 reactors, 88.5 GWe gross, 40 in China	710 GW installed in 2035 (IEA Net Zero)	Zirconium cladding tube, Alloy 690 tube
Installed electrolysis	More than 4 GW (2025)	Committed 2030 output 4.3 Mt of hydrogen	Nickel strip and mesh
Solid oxide fuel cells, Bloom Energy	1 GW a year, doubling to 2 GW by end 2026	Driven by data centre power	Ferritic chromium steel strip

Sources: Siemens Energy capital markets day 2025, World Nuclear Association (2026), IEA World Energy Outlook 2025 and Global Hydrogen Review 2026, Bloom Energy results (Bloom Energy, 2026a; International Energy Agency, 2025h, 2026m, 2026n; Siemens Energy, 2025; World Nuclear Association, 2026b).

Hydrogen shows the widest gap between political ambition and the orders that reach equipment suppliers. Global hydrogen demand surpassed 100 Mt in 2025, but low emissions production was still only about 1 Mt, and the announced 2030 pipeline shrank to 27 Mt (International Energy Agency, 2026l, 2026o, 2026p). thyssenkrupp nucera, the only named European buyer of nickel electrode material in the company universe, saw green hydrogen orders fall from EUR 356 million to EUR 26 million in fiscal 2024/25 and abandoned solid oxide stack mass production in August 2026 (thyssenkrupp nucera, 2025, 2026).

5.3.2 Key Applications and Opportunities for the Company

The company's benchmark portfolio covers superalloys for turbines, Nickel 201 strip for alkaline electrolyser cathodes, ferritic interconnect alloys for solid oxide cells, Alloy 690 and zirconium for nuclear plants, and corrosion alloys for flue gas cleaning and waste to energy boilers.

Table 8: The company's energy applications, outlook to 2035

Outlook	Applications	Materials and product forms
Grows strongly	Gas turbine hot section parts and repair	Superalloy bar, billet, welding wire
Grows	Solid oxide fuel cell interconnects for data centre power	Ferritic chromium strip
Grows from 2030	Nuclear steam generators and fuel	Alloy 690 tube stock, zirconium strip and tube
Uncertain	Alkaline and solid oxide electrolysers	Nickel strip and mesh, interconnect strip
Stable	Flue gas desulphurisation, waste to energy overlay	Plate, welding consumables

Source: VDM Metals application benchmark, outlook is the team's assessment.

The nuclear pull is real but late, as the United States government framework of October 2025 covers at least USD 80 billion of AP1000 reactors, while Alleima is today the only confirmed supplier of Alloy 690 steam generator tube from Sandviken (Alleima AB, 2026a; Westinghouse Electric, 2025). Bloom Energy is the fastest growing interconnect buyer, with revenue of USD 2,024 million in 2025 and capacity doubling by the end of 2026, although it names neither its alloy grade nor its mill (Bloom Energy, 2026a, 2026b).

What the company should do:

- Qualify superalloy bar and billet with Siemens Energy and its casting and forging suppliers while capacity is sold out to 2028.
- Offer ferritic interconnect strip to Bloom Energy and other solid oxide fuel cell makers serving data centres.
- Prepare Alloy 690 and zirconium qualifications for European and American nuclear programmes that order from about 2028 onward.
- Treat electrolysers as an option, released for investment only against funded projects and signed purchase volumes.

5.4 Aerospace

Aerospace is the company's strongest growth market, because engine production, aftermarket services and defence demand all rise while qualified superalloy capacity stays short. Nickel and cobalt superalloys for jet engines are the core products, and ceramic matrix composites in the hottest engine sections are the main long term substitution risk.

5.4.1 Fleet, Engines and Services to 2035

Airbus expects the passenger fleet to grow from 23,310 aircraft at the end of 2025 to 45,550 in 2045, which requires 42,060 new passenger aircraft, and Boeing expects 43,625 deliveries over the same period on a wider fleet definition (Airbus, 2026d, 2026e; Boeing, 2026c). Engine output grows faster than aircraft output, as CFM delivered 1,802 LEAP engines in 2025, up 28 percent, and Rolls-Royce still describes parts availability as constrained (Safran, 2026a).

Table 9: Aerospace market indicators, today and 2035

Indicator	Today	2035	Source basis
Passenger fleet, aircraft of 100 seats or more	23,310 (2025)	About 32,600 (team interpolation)	Airbus GMF 2026
Commercial fleet, wider definition	About 28,000 (2025)	About 38,000	Boeing CMO 2026
New generation share of fleet	39% (2025)	82%	Airbus
Aviation services market	USD 209 bn (2026)	USD 422 bn in 2045, 3.8% a year	Airbus GSF 2026
Cobalt used in aviation	11,499 t (2025)	Grows with fleet and engine output	Cobalt Institute
World military expenditure	USD 2,887 bn (2025), Europe +14%	Rising in Europe	SIPRI

Sources: Airbus (2026), Boeing (2026), Cobalt Institute (2026), SIPRI (2026) (Airbus, 2026d, 2026f, 2026g; Boeing, 2026d; Cobalt Institute, 2026g; Stockholm International Peace Research Institute, 2026). The 2035 passenger fleet is a team estimate on a constant growth path and not an Airbus figure.

Services are the profit pool, since Airbus says that over 85 percent of off wing maintenance cost is driven by materials, and GE Aerospace took 75 percent of its commercial engine revenue from services in 2025 (Airbus, 2026g; GE Aerospace, 2026). Qualified capacity remains concentrated in the United States, the United Kingdom, France and Germany, while deliveries move toward Asia, so the buyers of bar and billet remain Western engine makers and forgers.

5.4.2 Key Applications and Opportunities for the Company

The company sells nickel superalloy bar, sheet and welding wire for hot section parts such as discs, shafts, casings and combustors, and cobalt alloy sheet for combustors and wear parts. Its direct buyers are engine makers and their forgers rather than the airframers.

Table 10: The company's aerospace applications, outlook to 2035

Outlook	Applications	Materials and product forms
Grows strongly	Rotating and static hot section parts for LEAP, GE9X, Trent and military engines	Superalloy bar and billet, sheet
Grows	Engine repair and overhaul	Welding wire, repair alloys
Grows	Additively manufactured engine parts	Superalloy powder
Stable	Combustor and wear parts where nickel alloys and ceramics replace cobalt	Cobalt alloy sheet and plate

Source: VDM Metals application benchmark, outlook is the team's assessment.

Customers are investing in captive capacity, which defines the window for a new supplier. Safran committed EUR 150 million to a 30,000 t press at Gennevilliers that will forge about 14,000 superalloy and titanium parts a year from 2029 (Safran, 2026b). ATI earned USD 1,762.9 million from commercial jet engines in 2025 under long term agreements with GE Aerospace, Safran, Rolls-Royce and Pratt and Whitney, and quotes forging lead times of more than two years (ATI Inc., 2026c, 2026d). Daido Steel is building a JPY 36 billion high alloy melting and forging base in Japan, so the window for European capacity is open now but will narrow by 2030 (SteelOrbis, 2026).

What the company should do:

- Qualify superalloy bar and billet as an additional source for the Safran Gennevilliers press before its 2029 start.
- Use the current shortages to win long term agreements with MTU, Rolls-Royce and GE Aerospace forgers, because approval takes years.
- Grow welding wire and repair alloys for the maintenance market, which grows by 3.8 percent a year to 2045.
- Secure cobalt from more than one region and recover customer revert, because the Congolese quota limits alloy grade supply.

5.5 Chemical Processing

Chemical processing is the company's core market for corrosion resistant alloys, but European demand now depends on maintenance and replacement rather than on new plants, while new capacity is built in Asia. The strongest opportunity is equipment where a longer service life pays for a higher material price.

5.5.1 European Weakness and Asian Growth

Europe's chemical industry is shrinking in volume and running with spare capacity. EU27 chemical sales were about EUR 635 billion in 2024, but Europe's share of world sales fell from 21 percent in 2009 to 13 percent, while China's rose from 24 to 46 percent (European Chemical Industry Council, 2025a, 2025b, 2025c). The EU chemical production index fell 3.2 percent in 2025, and capacity utilisation was 75.0 percent in the third quarter of 2026 against a long term average of 81.3 percent (European Chemical Industry Council, 2026; Eurostat, 2026b).

Table 11: Chemical industry indicators

Indicator	Value	Implication for alloy demand
EU27 chemical sales, 2024	About EUR 635 bn	Large installed base needing maintenance
EU share of world chemical sales	13% (2024), from 21% (2009)	New build moves out of Europe
EU chemical production index, 2025	Down 3.2%	Fewer debottlenecking projects
EU capacity utilisation, Q3 2026	75.0% against 81.3% average	Closures, not expansions
German utilisation, 2025	72.5%	Weakest home market
Asia Pacific share of growth to 2035	Over 80% (BASF)	Export projects through fabricators

Sources: Cefic (2025, 2026), Eurostat, VCI, BASF Report 2025 (BASF, 2026a; European Chemical Industry Council, 2025a, 2025b, 2026; Eurostat, 2026b; Verband der Chemischen Industrie, 2026).

New capital follows growth to Asia, as BASF expects over 80 percent of chemical industry growth to be in Asia Pacific by 2035 and started its Zhanjiang Verbund site in November 2025, while it plans 2026 capital spending below depreciation and has closed several Ludwigshafen units (BASF, 2024, 2026a, 2026b; BASF SE, 2026).

5.5.2 Key Applications and Opportunities for the Company

The company's chemical portfolio covers Nickel 200, Alloy 59, C-276, 625, Alloy 31 and zirconium for acids, caustics, heat exchangers, vessels, pumps, valves and furnaces. The evidence from competitors shows that the segment trades volume for price, as Haynes cut chemical processing volume by 20.8 percent in its last standalone year while revenue stayed flat because the price per pound rose 26.7 percent (Haynes International, 2023a). Acerinox called chemical processing significantly weaker in 2025 and expects it to stay weak through 2026 (Acerinox, 2026d).

Table 12: The company's chemical processing applications, outlook to 2035

Outlook	Applications	Materials and product forms
Stable	Replacement heat exchangers, vessels and piping in European plants	Plate, sheet, tube stock, welding consumables
Grows	New plants in China, India and Southeast Asia via EPC contractors	Plate, clad plate, zirconium
Grows	Battery material and hydrometallurgy plants	Plate, welding consumables
Falls	New build in European base chemicals	Plate

Source: VDM Metals application benchmark, outlook is the team's assessment.

The buyers are fabricators and equipment makers, while plant operators such as BASF set the specification, and Alleima is investing SEK 220 million in finishing at Mehsana in India to serve the same Asian demand (Alleima AB, 2026c). Where stainless steel, linings or other materials meet the duty economically, the company should accept a smaller market rather than assume universal substitution by nickel alloys.

What the company should do:

- Defend European replacement demand with plate and matching welding consumables sold through fabricators.
- Qualify with EPC contractors and fabricators that build Asian plants, starting in India and Southeast Asia.
- Sell on documented life cycle cost, with corrosion test data, rather than on the price per tonne.

5.6 Electronics and Electrical Engineering

Electronics and electrical engineering offers growth in precision materials for sensors, shielding, glass seals, display masks and resistance heating, while data centres raise demand for copper power distribution. The company's role is in selected components, so semiconductor sales growth is an indicator of activity rather than a multiplier for alloy volume.

5.6.1 Semiconductors, Data Centres and Displays

Semiconductor sales reached USD 795.6 billion in 2025, up 26.2 percent, and WSTS forecasts USD 1.51 trillion for 2026, driven mainly by memory (World Semiconductor Trade Statistics, 2026a, 2026b). Equipment spending is the closer indicator for metals, and SEMI expects 300 mm fab equipment spending of USD 149 billion in 2026 and USD 227 billion in 2029 (Evertiq, 2026; SEMI, 2026).

Table 13: Electronics indicators relevant to the company

Indicator	Value	Relevance
World semiconductor sales	USD 795.6 bn (2025), USD 1.51 tn (2026 forecast)	Activity, not metal volume
300 mm fab equipment spending	USD 149 bn (2026), USD 227 bn (2029)	Gas delivery and chamber parts in high purity alloys
Copper busbar in a 1 MW rack at 54 VDC	Up to 200 kg	Copper and copper alloy strip and bar
German electrical and electronics value added	EUR 112.2 bn (2023), real output flat in 2025	Home market for sensor and seal alloys
DNP metal mask capacity	Doubled by the 8th generation Kurosaki line	FeNi36 (invar) ultra thin foil

Sources: WSTS (2026), SEMI (2026), NVIDIA (2025), OECD STAN and ZVEI, Dai Nippon Printing (Dai Nippon Printing, 2024a; Evertiq, 2026; NVIDIA, 2025; OECD, 2026; World Semiconductor Trade Statistics, 2026a; ZVEI, 2026).

Data centres change the power architecture, since NVIDIA plans 800 VDC distribution with its Kyber racks in 2027 and names Infineon, Vertiv, Eaton and Schneider Electric among its partners (NVIDIA, 2025). Vertiv more than doubled its switchgear and busbar capacity and raised its copper hedge to 12,946 t by September 2026, and Infineon opened its EUR 5 billion Smart Power Fab in Dresden in July 2026 (Infineon Technologies, 2026; Vertiv, 2026).

5.6.2 Key Applications and Opportunities for the Company

The benchmark portfolio lists expansion alloys 42 and 29 18, soft magnetic grades and resistance alloys, used in glass and ceramic seals, shielding, toroidal cores, current measurement and heating elements, plus FeNi36 for display masks.

Table 14: The company's electronics applications, outlook to 2035

Outlook	Applications	Materials and product forms
Grows	OLED fine metal masks for IT panels	FeNi36 ultra thin foil, through mask makers
Grows	Current sensors and shielding in power electronics and data centres	Soft magnetic FeNi strip
Grows	Fab gas delivery and chamber components	High purity nickel alloy and stainless strip and tube stock
Stable	Glass seals, lead frames, resistance heating	FeNi42 strip, resistance wire

Source: VDM Metals application benchmark, outlook is the team's assessment.

The display niche carries a patent barrier, because Dai Nippon Printing holds the world's top share of OLED metal masks and told investors that its invar patents will not expire for the foreseeable future (Dai Nippon Printing, 2024b; Display Daily, n.d.). The fab equipment filings of TSMC, Applied Materials and Lam Research name no alloy, so qualification must run through their component suppliers rather than through the toolmakers themselves.

What the company should do:

- Partner with a mask maker such as DNP to qualify FeNi36 ultra thin foil, rather than compete with protected processes.
- Grow soft magnetic and expansion alloy strip with power electronics makers such as Infineon and their stamping suppliers.
- Qualify high purity alloy strip and tube stock with second tier suppliers of fab gas systems in Germany, Taiwan and the United States.

5.7 Medical Technology

Medical technology is a steady market for the company, driven by ageing populations rather than by fast growth, and its value is concentrated in a few implant makers. The best fit is cobalt chromium and zirconium bar and powder for joint implants, followed by stainless wire for instruments and leads.

5.7.1 Procedures, Ageing and Prices

Joint replacement is the most directly measurable driver of implant metal demand. In 2023 the OECD average was 198 hip and 156 knee replacements per 100,000 people, and rates rose over the decade in nearly every OECD country, with Germany, Switzerland, Australia, Finland and Denmark among the highest (OECD, 2025a). Europe's medical

technology market is estimated at about EUR 173 billion in 2025, with spending growth of 1.8 percent, and the United States accounts for 46.4 percent of global device sales (MedTech Europe, 2025, 2026).

Table 15: Medical technology indicators

Indicator	Value	Relevance
Hip replacements per 100,000 people, OECD average	198 (2023)	Volume driver for CoCr and zirconium bar
Knee replacements per 100,000 people, OECD average	156 (2023)	Volume driver for CoCr bar and powder
European medtech market	About EUR 173 bn (2025)	Devices and diagnostics, not metal
Stryker orthopaedics revenue	USD 9,469 m (2025)	Largest implant buyer
Zimmer Biomet knees and hips	USD 3,322 m and USD 2,094 m (2025)	Names cobalt chrome as a purchased commodity
China volume based procurement	Implant prices cut by about 82% (2022)	Volume without margin

Sources: OECD Health at a Glance 2025, MedTech Europe, company filings, National Healthcare Security Administration (MedTech Europe, 2026; National Healthcare Security Administration, 2022; OECD, 2025a; Stryker Corporation, 2026; Zimmer Biomet Holdings, Inc., 2026a, 2026b).

Prices are under pressure in China, where central purchasing cut the prices of artificial hips and knees by about 82 percent, so the profitable volume for a European supplier remains in Europe and the United States (National Healthcare Security Administration, 2022).

5.7.2 Key Applications and Opportunities for the Company

Implant makers buy cobalt chromium, titanium and zirconium in bar and increasingly in powder for additively manufactured implants. Zimmer Biomet names cobalt chrome among its purchased commodities, and the FDA approved an additively manufactured cobalt chromium talus implant in 2023, which confirms that the alloy remains accepted (U.S. Food and Drug Administration, 2023; Zimmer Biomet Holdings, Inc., 2026b). Smith & Nephew's oxidized zirconium contains virtually no nickel, cobalt or chromium and shows 94.1 percent survivorship at 20 years, which makes it both a substitution threat to cobalt chromium and a zirconium opportunity (Smith & Nephew plc, 2026).

Table 16: The company's medical applications, outlook to 2035

Outlook	Applications	Materials and product forms
Grows	Hip and knee implants in Europe and the United States	CoCr and zirconium bar
Grows	Additively manufactured implants	CoCr powder
Stable	Surgical instruments, implantable leads, imaging equipment	Stainless wire, nickel alloy strip
Falls	Implants for China under central purchasing	CoCr bar

Source: VDM Metals application benchmark, outlook is the team's assessment.

The alloy market is small and cyclical for suppliers, as Carpenter's medical sales fell to USD 278.4 million in fiscal 2026 because customers reduced inventory (Carpenter Technology, 2026a). The buyers are concentrated, with Stryker, Zimmer Biomet, Smith & Nephew and the orthopaedics business that Johnson & Johnson plans to separate accounting for most of the market (Johnson & Johnson, 2025).

What the company should do:

- Focus on CoCr and zirconium bar for implant makers, with Germany, Switzerland and Austria as home markets.
- Develop medical grade CoCr powder for additively manufactured implants under ISO 13485 quality management.
- Approach the separated Johnson & Johnson orthopaedics business early, because a new organisation reviews its supplier base.

6. Global Market Dynamics and Trade

The Trade dimension connects the supply base of chapter 4 to the industries of chapter 5, and it shows that the company competes in a small number of well defined product corridors rather than in a global commodity market. A country can export raw material while importing refined metal or finished alloy, so production leadership does not establish access to the grades and forms a buyer needs. All figures below come from UN Comtrade at product code level and are stated in current US dollars for 2024 unless noted.

6.1 Germany Leads the Trade in Qualified Nickel Products

Germany is the world's largest exporter of nickel plate, sheet, strip and foil and the second largest exporter of nickel bars, rods and wire, which places the company's home base at the centre of the qualified product trade. German exports of the three nickel mill product codes reached USD 1.42 billion in 2024, against USD 0.97 billion in 2019, an increase of 47 percent (United Nations Statistics Division, 2026b, 2026c, 2026d).

Table 17: Exports of nickel mill products and related codes by leading country, 2024

HS code and product	Leading exporters (USD million)	Germany net trade (USD million)
7505 Nickel bars, rods, profiles, wire	United States 1,400, Germany 638, United Kingdom 402, Italy 388, China 299	+79
7506 Nickel plate, sheet, strip, foil	Germany 578, United States 482, Japan 274, China 85, United Kingdom 59	+414
7507 Nickel tubes and fittings	United States 324, Germany 202, China 195, United Kingdom 168, Italy 141	+105
7502 Unwrought nickel	China 2,068, Indonesia 862, United Kingdom 768, Japan 404, United States 339	−736
8105 Cobalt and articles	United States 418, China 266, United Kingdom 218, Germany 149, Japan 96	+19
8311 Welding wire and electrodes	China 952, United States 380, Korea 196, Germany 160, Japan 148	+63

Source: UN Comtrade (2026), reporter based data for Germany, the United States, China, Japan, Korea, India, France, the United Kingdom, Italy and Indonesia (United Nations Statistics Division, 2026a, 2026b, 2026c, 2026d, 2026e, 2026f). Team calculation of net trade as exports minus imports.

The pattern is clear, because Germany imports raw nickel worth USD 736 million more than it exports, but exports more nickel mill products than it imports in every product code. Value is created by conversion in Germany, and the export unit value of nickel plate and strip near USD 32,500 per tonne is about twice the exchange price of the metal, which confirms that buyers pay for qualification and form rather than for nickel content (United Nations, 2026).

6.2 Corridors and Dependencies

The main German customers for nickel bar and wire are the United Kingdom and the United States, each with about USD 111 million in 2024, followed by Italy and France, which mirrors the aerospace engine corridors of chapter 5.4 (United Nations Statistics Division, 2026b). Nickel plate and strip goes mainly to Austria, Italy, China and the United States, which reflects the chemical, oil and gas and electronics fabricators in those countries (United Nations Statistics Division, 2026c). China is a net importer of cobalt products worth about USD 2.9 billion in 2024, which shows that the battery chain, not the alloy industry, sets the cobalt market (United Nations Statistics Division, 2026e).

Trade in finished equipment confirms in which countries the company's direct buyers sit. The United Kingdom exported USD 39.8 billion of turbojets and gas turbines in 2024 and the United States USD 14.1 billion, which places the engine makers that buy superalloy bar in those two countries and in France and Germany (United Nations Statistics Division, 2026g). The OECD input output tables add the industrial view, as world intermediate use of non ferrous metals reached about USD 249 billion in fabricated metal products, USD 220 billion in electrical equipment and USD 109 billion in motor vehicles in 2022 (OECD, 2025b, 2025c, 2025d).

6.3 Implications for the Company

Trade data supports three conclusions for the company's sales and sourcing strategy. First, the qualified corridors to the United Kingdom, the United States and France are aerospace corridors and justify direct key account management with engine makers and forgers. Second, the European fabricator corridors to Austria and Italy carry the chemical and oil and gas business and are best served through stock and distribution. Third, raw material dependence on unwrought nickel and cobalt calls for multi source contracts, because Germany's trade surplus exists only after conversion, which is also the stage at which the scenarios in chapter 7 measure demand.

7. Market Scenarios to 2035

The Outlook dimension turns the sector evidence into a common yardstick, so that industries measured in different units can be compared on one scale. The scenarios are a sensitivity analysis of demand for the specialty metal applications identified in chapter 5, not a forecast of industry revenue, metal prices or company sales, and no probabilities are assigned.

7.1 Market Model and Scenario Assumptions

Demand in each industry is indexed to 100 in 2025 and compounded for ten years with the formula I(2035) = 100 × (1 + g)^10, where g is the annual growth rate of relevant alloy demand. Each rate is a team assumption that is anchored to the strongest indicator in chapter 5 and adjusted for material intensity and substitution, as Table 18 shows. Metal prices and exchange rates are held constant, so the indices measure volume and mix rather than value.

Table 18: Scenario assumptions and their evidence anchors

Industry	Downside / base / upside, % a year	Evidence anchor for the base case
Aerospace	2.0 / 4.5 / 6.0	Passenger fleet about 3.4% a year, engine output and services growing faster (services 3.8% a year)
Energy and environmental technology	0.5 / 4.0 / 6.5	Gas turbine capacity up from 17 GW to above 30 GW a year, nuclear from about 2030, hydrogen only in the upside
Electronics and electrical engineering	0.0 / 3.0 / 5.0	300 mm equipment spending to USD 227 bn in 2029, flat German real output
Medical technology	1.0 / 2.5 / 4.0	Rising joint replacement rates, European spending growth of 1.8% in 2025
Chemical processing	−1.0 / 1.0 / 2.5	EU utilisation of 75%, growth concentrated in Asia
Oil and gas	−1.0 / 0.5 / 1.5	Upstream investment flat near USD 550 bn, gas up, oil down
Automotive	−3.0 / −1.5 / 0.0	Electric share of car sales rising from 25% to about 50% by 2035

Source: team assumptions based on the evidence in chapter 5 (Airbus, 2026g; European Chemical Industry Council, 2026; International Energy Agency, 2026h, 2026r; MedTech Europe, 2026; SEMI, 2026; Siemens Energy, 2025).

The downside case assumes delayed projects, weak replacement spending and faster substitution, for example by ceramic matrix composites in engines or lithium iron phosphate in batteries. The base case assumes steady replacement demand, gradual electrification and the engine and turbine ramp up that manufacturers have announced. The upside case adds faster adoption of qualified energy components, including electrolysers and solid oxide fuel cells, and a stronger European defence build up.

7.2 Results by Industry, Product, and Region

Table 19 shows that aerospace and energy technology dominate the outlook, while automotive is the only industry that shrinks in the base case.

Table 19: Demand index 2035 by industry (2025 = 100)

Industry	Downside	Base	Upside	Main product forms
Aerospace	122	155	179	Superalloy bar and billet, sheet, wire, powder
Energy and environmental technology	105	148	188	Superalloy bar, interconnect strip, Alloy 690, zirconium
Electronics and electrical engineering	100	134	163	FeNi strip and foil, high purity strip
Medical technology	110	128	148	CoCr and zirconium bar, CoCr powder
Chemical processing	90	110	128	Plate, sheet, welding consumables
Oil and gas	90	105	116	Strip, plate, bar, welding wire
Automotive	74	86	100	Heat resistant and soft magnetic strip

Source: team calculation, I(2035) = 100 × (1 + g)^10 with the rates of Table 18, rounded.

Two results of the scenario analysis matter most for the strategy of the company. Aerospace and energy technology together are the only industries that grow by more than 40 percent in the base case, and both depend on the same product family of vacuum melted superalloy bar and billet, so one capacity decision serves two markets. The spread

between downside and upside is widest in energy technology, which argues for staged investment that follows signed orders. Regionally the growth sits in European and American engine and turbine supply chains, in Asian chemical construction and in European and American implant markets, which is the geography that chapter 8 turns into priorities.

8. Strategic Recommendations

Chapter 8 integrates the four dimensions of the framework and answers the central question of where supply, trade and demand create the most attractive opportunities for the company through 2035. Each opportunity combines a material, a product form, an application, an industry and a region, and it is assessed on the eight criteria introduced in chapter 3.

8.1 Priority Products, Industries, Regions, and Customers

Table 20 rates eleven opportunities from the sector analysis and ranks them in three tiers. Ratings are qualitative team judgements based on the evidence cited in chapters 4 to 7, where high means attractive for the company, so that low competitive intensity and high entry barriers for others both score high.

Table 20: Opportunity matrix, integration of Supply, Trade, Demand and Outlook

Opportunity (material, form, application, region)	Market	Growth	Fit	Competition	Technology	Customers	Barriers	Differentiation	Tier
Superalloy bar and billet for jet engines, Europe and United States	H	H	H	M	H	H	H	H	1
Superalloy bar for gas turbines, Europe and United States	H	H	H	M	H	H	H	H	1
Repair wire and superalloy powder for engines and turbines	M	H	H	M	H	H	M	H	1
Corrosion alloy plate and welding consumables for chemical plants, Europe and Asia	H	L	H	L	M	M	M	M	2
FeNi strip and foil for sensors and OLED masks, Europe and East Asia	M	M	H	M	H	M	H	H	2
CoCr and zirconium bar and powder for implants, Europe and United States	M	M	H	M	M	H	H	M	2
Sour service strip, bar and welding wire, Americas and Africa	M	L	H	M	M	M	H	M	2
Ferritic interconnect	L	H	M	M	H	M	M	H	3

Opportunity (material, form, application, region)	Market	Growth	Fit	Competition	Technology	Customers	Barriers	Differentiation	Tier
strip for fuel cells, United States									
Alloy 690 and zirconium for nuclear plants	M	M	M	H	M	H	H	M	3
Nickel strip and mesh for electrolysers	L	M	M	L	M	L	L	M	3
Heat resistant alloys for combustion engines	M	L	H	M	L	M	M	L	Harvest

Source: team assessment. H = high, M = medium, L = low attractiveness for the company. Tier 1 = invest to win, tier 2 = defend and grow selectively, tier 3 = option gated by customer milestones.

The first priority is vacuum melted superalloy bar, billet and repair wire for jet engines and gas turbines, sold to engine makers and their forgers in Europe and the United States, because it combines the highest growth with the highest entry barriers. Siemens Energy's capacity is sold out to 2028, Safran's new press starts in 2029, and ATI quotes forging lead times beyond two years, so qualified capacity is the scarce resource (ATI Inc., 2026d; Safran, 2026b; Siemens Energy, 2025).

The second tier protects and grows the existing franchises where the company's approvals, not volume, create value. Chemical plate and welding consumables remain the largest business but must follow construction to Asia through EPC contractors, while precision FeNi strip, medical CoCr and zirconium bar and sour service products grow selectively with named buyers such as Infineon, DNP, Zimmer Biomet, Stryker, BUTTING and TechnipFMC.

The third tier consists of options that are released for investment only after customer milestones. Fuel cell interconnect strip, nuclear Alloy 690 and zirconium, and electrolyser nickel each have credible demand but uncertain timing, and thyssenkrupp nucera's collapse in green hydrogen orders shows why the company should not build capacity ahead of signed volumes (thyssenkrupp nucera, 2025).

8.2 Recommended Actions and Implementation Roadmap

The roadmap sequences five initiatives over three horizons, so that each investment is released only after the previous step has proven demand.

Table 21: Implementation roadmap 2026 to 2035

Initiative	2026 to 2028: secure and qualify	2029 to 2031: scale	2032 to 2035: lead	KPI
1. Superalloy engine and turbine programme	Qualify bar and billet with Safran, MTU, Rolls-Royce and Siemens Energy suppliers, sign long term agreements	Add remelt and finishing capacity against signed volumes	Become second European source for hot section bar	Approved part numbers, LTA volume, share of sales
2. Repair and additive manufacturing	Launch repair wire range and qualify powder	Supply MRO shops in Europe and Asia	Powder for serial printed parts	Repair revenue, powder approvals
3. Chemical and energy fabricator network	Key accounts with EPC contractors for India and Southeast Asia, life cycle cost data	Stock and distribution in Asia	Share of new build projects	Order intake outside Europe
4. Precision and medical materials	Partner with a mask maker and sensor makers, ISO 13485 for CoCr	Grow FeNi foil and CoCr powder	Leading European source	Sample to approval conversion
5. Raw material resilience	Multi source cobalt and class 1 nickel, revert contracts, surcharge clauses	Closed loop scrap with aerospace customers	Low carbon product offer	Revert share, cost volatility

Source: team recommendation based on chapters 4 to 7.

Execution of the roadmap follows three governance rules that protect the company's capital. Capacity is added only where qualified demand shows a constraint and the investment earns its hurdle rate in the downside scenario of chapter 7. Commercial resources follow the tiers of Table 20, which means fewer resources for diesel engine parts and European new build chemicals. Progress is reviewed quarterly against approvals, repeat orders, contribution margin per bottleneck hour and on time delivery.

The main risks to this strategy are known, and each of them can be managed with a specific response. Cobalt export quotas and Indonesian nickel policy are addressed by multi sourcing and revert recovery, the substitution of nickel alloys by ceramic matrix composites and titanium aluminide by early participation in customer development programmes, and new Asian capacity at Daido and Chinese producers by fast qualification before 2030 (SteelOrbis, 2026).

9. Conclusion

The report asked how the production, availability, trade and industrial uses of selected metals shape opportunities in high performance materials through 2035, and the answer is that value lies in conversion and qualification rather than in metal volume. Raw materials are plentiful in nickel and policy constrained in cobalt and tungsten, Germany earns its trade surplus only after conversion, and the industries that pay most for qualified conversion are aerospace and energy technology, whose base case demand rises by about half by 2035.

The company should therefore concentrate investment on vacuum melted superalloy bar, billet, repair wire and powder for engine makers and turbine builders in Europe and the United States, defend its chemical, precision, medical and sour service franchises with targeted sales, and treat fuel cells, nuclear and electrolysers as options released by customer milestones. The central limitation of this study is the absence of internal sales, margin, approval and capacity data, so the next step is to validate the priorities against that evidence and convert the strongest opportunities into qualified orders.

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Appendices: Data, Sources, Calculations, and Assumptions

A1 Definitions and the stage ladder

The report uses Mt for million tonnes, kt for thousand tonnes, t for metric tonnes, GW for gigawatts and bcm for billion cubic metres, and all prices are in current US dollars or euros of the year stated. The stage ladder separates five rungs that are never added together: mine output of contained metal, refined metal or intermediate such as nickel pig iron or cobalt hydroxide, alloy melt, mill product such as strip, sheet, plate, bar and wire, and finished component. Contained metal is kept apart from gross weight, actual output from capacity, and commodity benchmarks such as the World Bank Pink Sheet from the selling price of qualified alloy products.

A2 Reproducible scenario calculations

Each 2035 index uses the formula I(2035) = 100 × (1 + g)^10 with 2025 as the base year, rounded to the nearest integer. For example, the aerospace base case is 100 × 1.045^10 = 155.3, rounded to 155, and the automotive base case is 100 × 0.985^10 = 86.0, rounded to 86. The rates in Table 18 are team assumptions anchored to cited evidence and are not third party alloy forecasts, and no company shipment weights or probabilities are applied.

A3 Data sources, product codes and reproducibility

All structured data were pulled with the project's fetch script, which stores raw files, writes clean tables and logs every query with its date and URL in the source register of 1,030 entries. UN Comtrade data cover the product codes HS 7502, 7504, 7505, 7506, 7507, 8105, 8109, 7219, 7220, 7223, 7409, 7606, 8311 and 8411 for ten reporting countries from 2019 to 2024, and Eurostat PRODCOM data cover fifteen nickel, cobalt, stainless, copper and aluminium product codes for the EU and five producer countries. USGS Mineral Commodity Summaries 2026, the World Bank Pink Sheet, OECD STAN and ICIO, UNIDO INDSTAT for Germany and the filings of 61 companies complete the evidence base.

A4 Information needed before investment

The market findings must be tested against internal data before they become a business plan. The company needs its actual shipments and net of metal sales by application, conversion margins, production yields and bottleneck hours, its approved customer, grade and form combinations, qualification lead times and firm order coverage. The legal group structure should be validated when comparing competitors, so that VDM Metals and Haynes are counted once inside Acerinox, and scenario rates should be replaced with customer level quantities as soon as these become available.